

Chemical Kinetics

Question1

$R \longrightarrow P$ is a first order reaction. The concentration of R changed from 0.04 to 0.03 mol L^{-1} in 40 min . What is the average velocity of the reaction in $\text{mol L}^{-1} \text{ s}^{-1}$?

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Options:

A.

$$2.5 \times 10^{-4}$$

B.

$$4.167 \times 10^{-6}$$

C.

$$4.167 \times 10^6$$

D.

$$2.5 \times 10^{-5}$$

Answer: B

Solution:

The change in concentration, $\Delta[R]$ is the final concentration minus initial concentration.

$$\Delta[R] = 0.03 - 0.04 = -0.01 \text{ mol/L}$$

$$\begin{aligned} \text{Average velocity} &= -\frac{\Delta[R]}{\Delta t} = -\left(-\frac{0.01}{40}\right) \\ &= 0.00025 \text{ mol}^{-1} \text{ min}^{-1} \end{aligned}$$

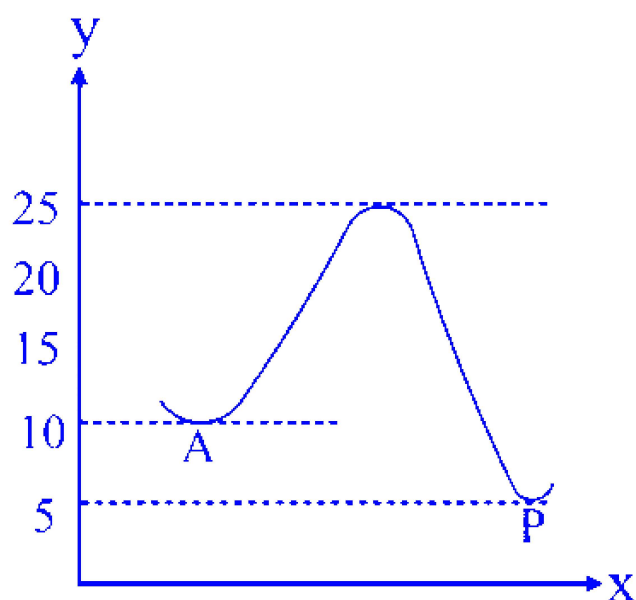
$$\text{or } 2.5 \times 10^{-4} \text{ mol}^{-1} \text{ L}^{-1} \text{ min}^{-1}$$

$$\ln \text{Mol}^{-1} \text{L}^{-1} \text{s}^{-1} = 4.167 \times 10^{-6} \text{mol}^{-1} \text{L}^{-1} \text{s}^{-1}$$

Question2

The following graph is obtained for a first order reaction ($A \rightarrow P$). The activation energy (E_a in kJmol^{-1}) and heat of reaction ($|\Delta H|$ in kJmol^{-1}) for this reaction are respectively

($x = \text{reaction coordinate}$; $y = E$ in kJmol^{-1})



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Options:

A.

5,15

B.

15,5

C.

25,5

D.

10,25

Answer: B

Solution:

From the graph,

Energy of reactant $A = 10 \text{ kJ/mol}$

Energy of product $P = 5 \text{ kJ/mol}$

Energy of transition state (peak) $= 25 \text{ kJ/mol}$.

Activation energy $E_a = \text{Energy of transition state} - \text{Energy of reactant}$

$$25 - 10 = 15 \text{ kJ/mol}$$

Heat of reaction $\Delta H = \text{Energy of product}$

$- \text{Energy of reactant}$

$$\Delta H = 5 - 10 = -5 \text{ kJ/mol}$$

$$|\Delta H| = 5 \text{ kJ/mol}$$

Question3

For a first order reaction, the ratio between the time taken to complete $\frac{3}{4}$ th of the reaction and time taken to complete half of the reaction is

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Options:

A.

2

B.

3

C.

1.5

D.

2.5

Answer: A

Solution:

For half completion $[A]_t = \frac{[A]_0}{2}$

$$t_{1/2} = \frac{2.303}{K} \log \frac{[A]_0}{[A]_{0/2}}$$

$$\text{Simplify } t_{1/2} = \frac{0.693}{K} \dots (i)$$

for 3/4 completion

$$[A]_t = \frac{1}{4}[A]_0$$

$$t_{3/4} = 2 \times t_{1/2} \dots (ii)$$

Taking ratio of Eqs. (i) and (ii),

$$\frac{t_{3/4}}{t_{1/2}} = \frac{2 \times t_{1/2}}{t_{1/2}} = 2 : 1$$

Question4

The following equation is obtained for a first order reaction at 300 K

$$\log_{10} \frac{k}{A} = 0.00174$$

What is the activation energy (in J mol^{-1}) of the reaction?

$$\left(R = 8314 \text{ J mol}^{-1} \text{ K}^{-1} \right)$$

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Options:

A.

10.0

B.

100.0

C.

0.1

D.

1.0

Answer: A

Solution:

The given equation is

$$\log_{10} \frac{k}{A} = 0.00174$$

Multiply 2.303 on both side

$$\ln \left(\frac{k}{A} \right) = 2.303 \times 0.00174$$

$$\Rightarrow \ln \left(\frac{k}{A} \right) = 0.00400722 \quad \dots (i)$$

The Arrhenius equation is

$$k = Ae^{-E_a/RT} \Rightarrow \frac{k}{A} = e^{-E_a/RT}$$

Taking natural log

$$\ln \left(\frac{k}{A} \right) = -\frac{E_a}{RT} \quad \dots (ii)$$

Compare Eqs. (i) and (ii)

$$-\frac{E_a}{RT} = 0.00400722$$

Substituting all values

$$E_a \simeq 10.0 \text{ J/mol}$$

Question5

$A \rightarrow B$ is a first order reaction. The concentration of A is decreased from $x \text{ mol L}^{-1}$ to $y \text{ mol L}^{-1}$ in 100 min . What is the average velocity of the reaction in $\text{mol L}^{-1} \text{ min}^{-1}$?

Options:

A.

$$\frac{|x-y|}{100}$$

B.

$$\frac{|y-x|^2}{100}$$

C.

$$\frac{100}{|x-y|}$$

D.

$$\frac{100}{|x+y|}$$

Answer: A

Solution:

We want to find the average velocity (rate) for the reaction as A goes to B when the amount of A drops from x to y mol/L in 100 minutes.

The change in the concentration of A is: $\Delta[A] = y - x$ mol/L. This tells us how much A has changed in value.

The average velocity means how much A changes per minute. We can find it by dividing the total change in A by the time taken: Average velocity = $\frac{\Delta[A]}{\Delta t}$ where $\Delta t = 100$ minutes.

So, Average velocity = $\frac{y-x}{100}$ mol L⁻¹ min⁻¹. If you need the answer as a positive value (since rates are usually positive), write it as $\frac{|x-y|}{100}$ mol L⁻¹ min⁻¹.

Question 6

In a first order reaction, the concentration of the reactant is reduced to $1/8$ of the initial concentration in 75 minutes. The $t_{1/2}$ of the reaction (in minutes) is ($\log 2 = 0.30$, $\log 3 = 0.47$, $\log 4 = 0.60$)

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Options:

A.

60.2

B.

50.2

C.

25.1

D.

75.1

Answer: C

Solution:

The concentration becomes $\frac{1}{8}$ of what it was at the start.

$$\frac{1}{8} = \left(\frac{1}{2}\right)^3$$

This means it takes 3 half-lives for the concentration to become $\frac{1}{8}$.

$$t_{1/2} = \frac{75}{3}$$

$$t_{1/2} = 25 \text{ minutes}$$

Question7

At T (K) the following equation is obtained for a first order reaction $\log \frac{k}{A} = -\frac{x}{T}$. The activation energy for this reaction is equal to (R = gas constant)

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Options:

A.

$$2.303 \times R$$

B.

$$\frac{2.303R}{x}$$

C.

$$\frac{x}{2.303R}$$

D.

$$\frac{1}{2.303 \times R}$$

Answer: A

Solution:

Given equation,

$$\log \left(\frac{K}{A} \right) = -\frac{x}{T} \quad \dots (i)$$

Using Arrhenius equation

$$k = Ae^{-E_a/RT}$$

Taking log on both side

$$\log k = \log A - \frac{E_a}{2.303RT}$$

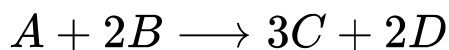
$$\log \left(\frac{k}{A} \right) = -\frac{E_a}{2.303RT} \quad \dots (ii)$$

Compare Eqs. (i) and (ii),

$$\frac{x}{T} = \frac{E_a}{2.303RT} \Rightarrow E_a = 2.303R \cdot x$$

Question8

Consider the reaction given below



If rate of disappearance of B is $x \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$, the ratio of rate of reaction and rate of appearance of C is

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Options:

A.

1 : 3

B.

3 : 1

C.

1 : 2

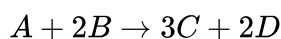
D.

2 : 1

Answer: A

Solution:

Given reaction is,



The rate of reaction is, $r = -\frac{1}{2} \frac{d[B]}{dt}$

Given, $\frac{d[B]}{dt} = -x \times 10^{-2}$

Thus, $r = \frac{1}{2} \times x \times 10^{-2} \quad \dots (i)$

Rate of appearance of C

$$\begin{aligned} &= \frac{1}{3} \frac{d[C]}{dt} = -\frac{1}{2} \frac{[dB]}{dt} \\ \frac{d[C]}{dt} &= \frac{3}{2} \times x \times 10^{-2} \quad \dots (ii) \end{aligned}$$

Taking the ratio of (i) and (ii)

$$\frac{\text{rate of reaction}}{\text{rate of appearance of } C} = 1 : 3$$

Question9

Activation energy for the hydrolysis of sucrose by acid is $X \text{ kJ mol}^{-1}$ whereas activation energy for the hydrolysis of sucrose by sucrase is $Y \text{ kJ mol}^{-1}$. X and Y respectively are

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Options:

A.

6.22, 2.15

B.

2.15, 6.22

C.

6.22, 6.22

D.

2.15, 2.15

Answer: A

Solution:

The activation energy for acid hydrolysis of sucrose is 6.22 kJ/mol. The activation energy for sucrose by sucrase is 2.15 kJ/mol.

Question10

$A \rightarrow$ products, is a first order reaction. The following data is obtained for this reaction at T (K). The value of $x : y$ is

Rate $\left(\text{molL}^{-1} \text{min}^{-1} \right)$	$[A]$
0.2	0.02M
0.4	x M
1.0	y M

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Options:

A.

1 : 5

B.

2 : 3

C.

5 : 2

D.

2 : 5

Answer: D

Solution:

For 1st order rate is, $\text{Rate} = k[A]$

$$\frac{\text{Rate}_1}{\text{Rate}_2} = \frac{[A_1]}{[A_2]}$$

For first two data points,

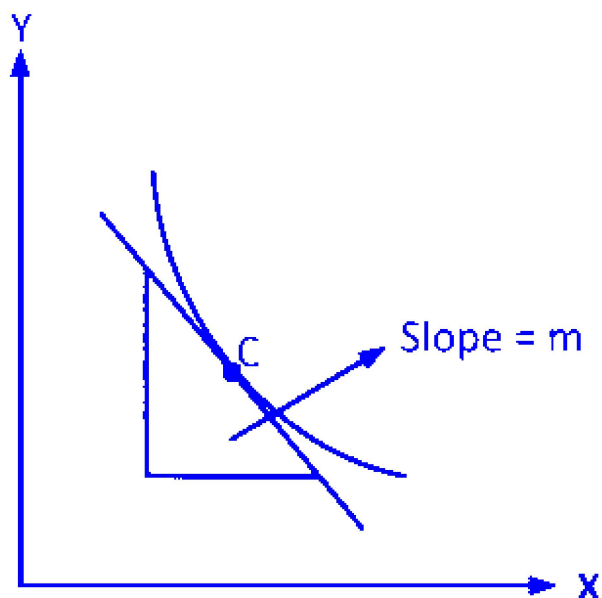
$$\begin{aligned} \frac{0.2}{0.4} &= \frac{0.02}{x} \\ \Rightarrow x &= 0.04 \text{ m} \end{aligned}$$

For second and third data point,

$$\begin{aligned} \frac{0.4}{1} &= \frac{0.04}{y} \Rightarrow y = 0.1 \text{M} \\ x : y &\Rightarrow 0.04 : 0.1 \Rightarrow 2 : 5 \end{aligned}$$

Question11

$A \rightarrow P$ is a first order reaction. The following graph is obtained for this reaction. (X-axis = time: Y-axis = conc. of A) The instantaneous rate of the reaction at point C is



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Options:

A. $\frac{1}{m}$

B. m

C. $2.303 m$

D. $\frac{1}{2.303m}$

Answer: B

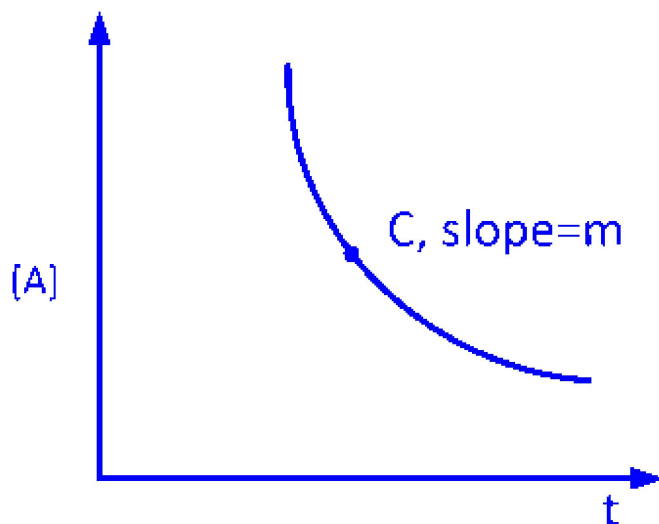
Solution:

The instantaneous 1st order reaction rate is given as

$$\text{rate} = \frac{-d[A]}{dt} \quad \dots (i)$$

On X -axis = time, Y -axis = concentration

From graph slope = $-m$



i.e., concentration is decreasing with time.

$$-m = \frac{d[A]}{dt} \text{ from graph} \quad \dots (ii)$$

Equating (i) and (ii)

$$\text{Rate of reaction} = m$$

Question12

The rate constant of a first order reaction was doubled when the temperature was increased from 300 to 310 K . What is its approximate activation energy (in kJmol^{-1})? ($R = 8.3\text{Jmol}^{-1} \text{ K}^{-1}$: $\log 2 = 0.3$)

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Options:

- A. 5.33
- B. 533.3
- C. 53333
- D. 53.33

Answer: D

Solution:

To find the approximate activation energy (E_a) for a first-order reaction where the rate constant doubles when the temperature changes from 300 K to 310 K, we can use the following relationship:

$$\log \left(\frac{K_2}{K_1} \right) = \frac{E_a}{2.303R} \frac{T_2 - T_1}{T_2 \times T_1}$$

For $T_1 = 300$ K, the rate constant is $K_1 = K$.

For $T_2 = 310$ K, the rate constant becomes $K_2 = 2K$.

By substituting the known values into the equation:

$$\log \left(\frac{2K}{K} \right) = \frac{E_a}{2.303 \times 8.314} \times \frac{310 - 300}{310 \times 300}$$

Simplifying and solving the equation:

$$\log 2 = \frac{E_a}{2.303 \times 8.3} \times \frac{10}{310 \times 300}$$

Given that $\log 2 = 0.3$, we rearrange to find E_a :

$$E_a = 5333 \text{ J/mol} = 53.33 \text{ kJ/mol}$$

Thus, the approximate activation energy is 53.33 kJ/mol.

Question13

Isomerisation of gaseous cyclobutene to butadiene is first order reaction. At T (K). The rate constant of reaction is $33 \times 10^{-4} \text{ s}^{-1}$. What is the time required (in min) to complete 90% of this reaction at the temperature? ($\log 2 = 0.3$)

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Options:

- A. 116.67
- B. 233.34
- C. 58.34
- D. 350.0

Answer: A

Solution:

For a first-order reaction, the integrated rate law is:

$$k = \frac{1}{t} \ln \frac{[A]_0}{[A]_t} \quad \dots(i)$$

Where:

k is the rate constant,

$[A]_0$ is the initial concentration,

$[A]_t$ is the concentration at time t .

Given:

$$k = 3.3 \times 10^{-4} \text{ s}^{-1},$$

$$[A]_0 = 100 \text{ (assuming initial concentration as 100 for calculation),}$$

$$[A]_t = 10 \text{ (since 90\% of the reaction is complete, only 10\% remains).}$$

Substitute these values into equation (i):

$$t = \frac{1}{k} \ln \frac{[A]_0}{[A]_t} = \frac{1}{3.3 \times 10^{-4}} \ln \left(\frac{100}{10} \right)$$

This simplifies to:

$$t = \frac{10^4 \times 2.303}{3.3} = 6977.53 \text{ s}$$

Converting seconds to minutes:

$$\frac{6977.5}{60} = 116.67 \text{ min}$$

So, the time required to complete 90% of the reaction is 116.67 minutes.

Question14

$A \rightarrow P$ is a zero order reaction. At 298 K the rate constant of the reaction is $1 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. Initial concentration of 'A' is 0.1 mol L^{-1} . What is the concentration of 'A' after 10 sec ?

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Options:

A. 0.09 mol L^{-1}

B. 0.099 mol L^{-1}

C. 0.087 mol L^{-1}

D. 0.011 mol L^{-1}

Answer: A

Solution:

For a zero-order reaction, the rate of reaction is independent of the concentration of the reactant. Given the rate constant $k = 1 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$, the initial concentration of A is $[A]_0 = 0.1 \text{ mol L}^{-1}$, and the time $t = 10$ seconds, we use the following formula to calculate the concentration of A after 10 seconds:

$$k = \frac{[A]_0 - [A]_t}{t}$$

Substituting the known values into the formula, we get:

$$1 \times 10^{-3} = \frac{0.1 - [A]_t}{10}$$

Solving for $[A]_t$:

$$0.01 = 0.1 - [A]_t$$

$$[A]_t = 0.1 - 0.01$$

$$[A]_t = 0.09 \text{ mol L}^{-1}$$

Therefore, the concentration of A after 10 seconds is 0.09 mol L^{-1} .

Question15

The rate constant of a first order reaction is $3.46 \times 10^{-2} \text{ s}^{-1}$ at 298 K . What is the rate constant of the reaction at 350 K if its activation energy is 50.1 kJ mol^{-1} , $\left(R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}\right) (\log 2 = 0.3010)$

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Options:

A. 0.592 s^{-1}

B. 0.692 s^{-1}

C. 0.792 s^{-1}

D. 0.892 s^{-1}

Answer: B

Solution:

To determine the rate constant of a reaction at a different temperature using the Arrhenius equation, follow this process:

The formula for calculating the change in the rate constant with temperature is:

$$\log \frac{K_{T_2}}{K_{T_1}} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Where:

K_{T_1} is the rate constant at temperature T_1 .

K_{T_2} is the rate constant at temperature T_2 .

E_a is the activation energy.

R is the universal gas constant.

Given Values:

Initial rate constant, $K_{T_1} = 3.46 \times 10^{-2} \text{ s}^{-1}$ at $T_1 = 298 \text{ K}$

Activation energy, $E_a = 50.1 \text{ kJ/mol} = 50.1 \times 10^3 \text{ J/mol}$

Universal gas constant, $R = 8.314 \text{ J/K} \cdot \text{mol}$

Temperature, $T_2 = 350 \text{ K}$

Calculation Steps:

Substitute the given values into the formula:

$$\log \frac{K_{T_2}}{3.46 \times 10^{-2}} = \frac{50.1 \times 10^3}{2.303 \times 8.314} \times \left(\frac{350 - 298}{298 \times 350} \right)$$

Evaluate the expression:

$$\log \frac{K_{T_2}}{3.46 \times 10^{-2}} = 1.30$$

Solve for K_{T_2} :

$$\frac{K_{T_2}}{3.46 \times 10^{-2}} = 10^{1.30} = 19.95$$

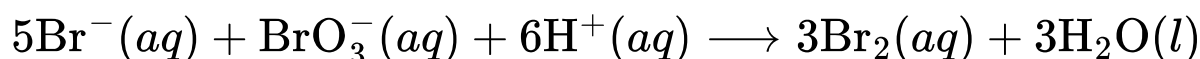
Calculate K_{T_2} :

$$K_{T_2} = 3.46 \times 10^{-2} \times 19.95 = 0.6903 \text{ s}^{-1}$$

Thus, the rate constant at 350 K is approximately 0.692 s^{-1} .

Question 16

At 298 K the value of $-\frac{\Delta[\text{Br}^-]}{\Delta t}$ for the reaction,



is $X \text{ mol L min}^{-1}$. What is the rate (in $\text{mol L}^{-1} \text{ min}^{-1}$) of this reaction?

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Options:

A. $5x$

B. x

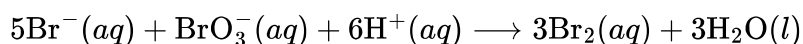
C. $\frac{x}{5}$

D. $-\frac{x}{5}$

Answer: C

Solution:

For the given reaction:



The rate of change for the concentration of Br^- ions is given as:

$$-\frac{\Delta[\text{Br}^-]}{\Delta t} = x \text{ mol L}^{-1} \text{ min}^{-1}$$

The overall rate of the reaction can be calculated by considering the stoichiometry of Br^- . Since 5 moles of Br^- are consumed per mole of reaction, the rate of the reaction is:

$$\begin{aligned} \frac{\Delta R}{\Delta t} &= -\frac{1}{5} \frac{\Delta[\text{Br}^-]}{\Delta t} \\ &= -\frac{1}{5} \times (-x) = \frac{x}{5} \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

Therefore, the rate of the reaction is $\frac{x}{5} \text{ mol L}^{-1} \text{ min}^{-1}$.

Question17

For a first order reaction the concentration of reactant was reduced from 0.03 mol L^{-1} to 0.02 mol L^{-1} in 25 min . What is its rate (in $\text{mol L}^{-1} \text{ s}^{-1}$)?

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Options:

A. 6.667×10^{-6}

B. 4×10^{-4}

C. 6.667×10^{-4}

D. 4×10^{-6}

Answer: A**Solution:**

The average rate of a first-order reaction is calculated using the formula:

$$\frac{\Delta R}{\Delta t} = - \left[\frac{R_2 - R_1}{t_2 - t_1} \right]$$

In this context:

R_1 is the initial concentration:

$$R_1 = 0.03 \text{ mol/L}$$

R_2 is the final concentration:

$$R_2 = 0.02 \text{ mol/L}$$

$t_2 - t_1$ is the time interval over which the change occurs, given as 25 minutes. To convert this time into seconds:

$$t_2 - t_1 = 25 \times 60 \text{ seconds}$$

By substituting these values into the rate formula, we obtain:

$$\frac{\Delta R}{\Delta t} = - \frac{0.02 - 0.03}{25 \times 60}$$

Calculating the above expression provides:

$$\frac{\Delta R}{\Delta t} = 6.667 \times 10^{-6} \text{ mol/L/s}$$

Question18

The first order reaction, $A(g) \rightarrow B(g) + 2C(g)$ occurs at 25°C . After 24 minutes the ratio of the concentration of products to the concentration of the reactant is 1 : 3 What is the half life is of the reaction (in min)? $\log 1.11 = 0.046$

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Options:

A. 150.5

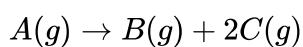
B. 142.2

C. 157.8

D. 15.78

Answer: C**Solution:**

For a first-order reaction,



After 24 minutes, the ratio of the concentration of the products to the reactant is 1 : 3.

The rate constant k for a first-order reaction is given by the formula:

$$k = \frac{2.303}{t} \log \left[\frac{a}{a-x} \right]$$

Where:

$t = 24$ minutes

$a = 4$ (initial concentration)

$a - x = 3$

Substituting these into the equation, we have:

$$k = \frac{2.303}{24} \log 1.11$$

Using $\log 1.11 = 0.046$, we get:

$$k = \frac{2.303}{24} \times 0.046 \approx 4.40 \times 10^{-3} \text{ min}^{-1}$$

The half-life ($t_{1/2}$) for a first-order reaction is calculated using:

$$t_{1/2} = \frac{0.693}{k}$$

Substitute $k = 4.40 \times 10^{-3} \text{ min}^{-1}$:

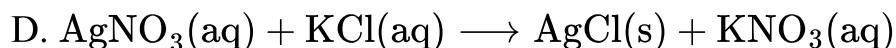
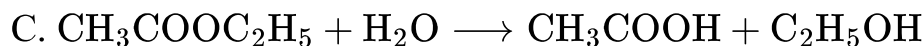
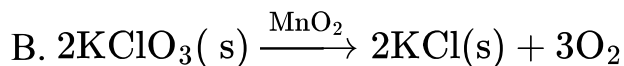
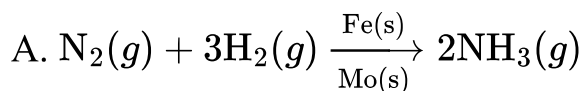
$$t_{1/2} = \frac{0.693}{4.40 \times 10^{-3}} \approx 157.8 \text{ minutes}$$

Question19

Identify the autocatalytic reaction from the following

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Options:

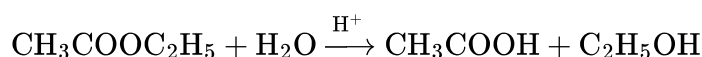


Answer: C

Solution:

A reaction is described as autocatalytic when one of its products also serves as a catalyst for the same reaction.

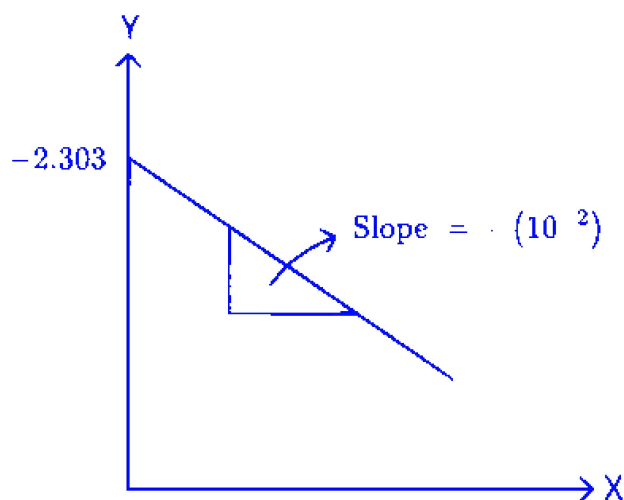
For the given reaction:



Here, CH_3COOH is both a product and a catalyst for the reaction, illustrating the autocatalytic nature.

Question20

At 298 K , for a first order resction ($A \rightarrow P$) the following graph is obtained. The rate constant (in s^{-1}) and initial concentration (in mol L^{-1}) of ' A ' are respectively (Y-axis = $\ln(a - x)$: X-axis = time in sec)



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Options:

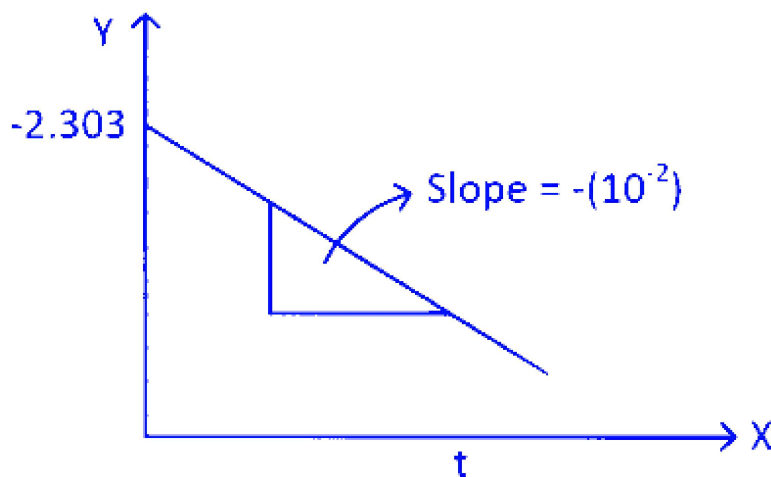
- A. 2.303, 10^{-1}
- B. 10^{-2} ; 2303
- C. 10^{-1} ; 10^{-2}
- D. 10^{-2} : 10^{-1}

Answer: D

Solution:

Given, first order reaction graph between $\ln(a - x)$ on Y -axis and t on X -axis.

According to first order reaction, $\ln(a - x)$



Rate constant is given by

$$k = \frac{\ln \frac{[a]}{[a-x]}}{t} \quad \dots (i)$$

$$\text{or } kt = \ln \frac{[a]}{[a-x]}$$

$$kt = \ln[a] - \ln[a-x]$$

$$\text{or } \ln[a-x] = \ln[a] - kt \quad \dots (ii)$$

Compare Eq. (ii) by straight line equation.

$$Y = \ln[a-x], x = t, m(\text{slope}) = -k$$

$$C(\text{intercept}) = \ln[a]$$

$$C(\text{intercept}) = \ln[a]$$

Therefore, slope = rate constant

$$= -(-10^{-2}) = 10^{-2} \quad \dots (iii)$$

Now initial concentration (time $t = 0$) Eq. (ii) becomes

$$\ln[a-x] = \ln[a]$$

$$\text{From graph at } t = 0, \ln[a-x] = -2.303$$

$$-2.303 = \ln[a]$$

$$\text{Initial concentration, } [a] = 10^{-1}$$

Question21

The rate constant of a reaction at 500 K and 700 K are 0.02 s^{-1} and 0.2 s^{-1} respectively. The activation energy of the reaction (in kJmol^{-1}) is $\left(R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}\right)$

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Options:

A. 66.90

B. 33.45

C. 22.30

D. 44.45

Answer: B

Solution:

Given,

$$k_1 = 0.02 \text{ s}^{-1}$$

$$k_2 = 0.2 \text{ s}^{-1}$$

$$E_d = ?$$

$$R = 8.3$$

$$T_1 = 500 \text{ K}$$

$$T_2 = 700 \text{ K}$$

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{0.2}{0.02} = -\frac{E_a}{8.3} \left[\frac{1}{700} - \frac{1}{500} \right]$$

$$\ln 10 = \frac{-E_a}{8.3} \left(\frac{500 - 700}{700 \times 500} \right)$$

$$2.302 \times 8.3 \times 700 \times 500 = 200E_a$$

$$\Rightarrow E_a = \frac{2.302 \times 8.3 \times 700 \times 500}{200}$$

$$= 33436 \text{ J/mol}$$

$$\approx 33.44 \text{ kJ/mol}$$

Question22

The time required for completion of 93.75% of a first order reaction is x minutes. The half-life of it (in minutes) is

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Options:

A. $x/8$

B. $x/2$

C. $x/4$

D. $x/3$

Answer: C

Solution:

Conc. at time, t , $A = A_0 - \frac{93.75}{100} A_0 = \frac{6.25}{100} A_0$

$A_0 \rightarrow$ initial conc. $k \rightarrow$ rate constant, time, $t = x$ min Using integrated rate law for 1st order reaction,

$$\begin{aligned} k &= \frac{2.303}{t} \log \frac{A_0}{A} \\ &= \frac{2.303}{x} \log \frac{100A_0}{6.25A_0} \end{aligned}$$

$$\begin{aligned} \text{So, } k &= \frac{2.303}{x} \log \frac{100}{6.25} \\ &= \frac{2.303 \times 1.20}{x} \\ \Rightarrow k &= \frac{2.303 \times 1.20}{x} \text{ min}^{-1} \end{aligned}$$

We know,

$$\begin{aligned} t_{1/2} &= \frac{0.693}{k} \\ &= \frac{0.693x}{2.303 \times 1.20} \end{aligned}$$

$$\text{So, } t_{1/2} = \frac{x}{4} \text{ min}$$

Question23

. The rate constant for a zero order reaction $A \longrightarrow$ products is $0.0030 \text{ mol L}^{-1} \text{ S}^{-1}$. How long it will take for

the initial concentration of A to fall from 0.10 M to 0.075 M ?

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Options:

- A. 10 s
- B. 20 s
- C. 8.33 s
- D. 1.33 s

Answer: C

Solution:

Given, rate constant, $k_0 = 0.003 \text{ mol L}^{-1} \text{ s}^{-1}$

Initial conc., $A_0 = 0.1 \text{ M}$

Final conc., $A = 0.075 \text{ M}$

Using integrated rate law for zeroth order,

$$A_0 - A = kt$$

$$0.1 - 0.075 = 0.003 \times t$$

$$t = \frac{0.025}{0.003} = 8.33 \text{ s}$$

Question 24

For a $A + B \rightarrow \text{products}$, the rate of the reaction is given by rate $= k[A][B]^2$. The units of rate constant (k) will be

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Options:

- A. $\text{mol L}^{-1} \text{ s}^{-1}$

B. L mol s^{-1}

C. $\text{mol}^2 \text{L}^{-2} \text{s}^{-1}$

D. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Answer: D

Solution:

Given:

$$\text{Rate} = k[A][B]^2$$

Step 1: Write dimensions of each term

- Rate of reaction has units:

$$\text{Rate} = \text{mol L}^{-1} \text{s}^{-1}$$

- $[A]$ and $[B]$ (concentrations) both have units:

$$\text{mol L}^{-1}$$

Step 2: Substitute into the rate equation

$$\text{mol L}^{-1} \text{s}^{-1} = k \times (\text{mol L}^{-1}) \times (\text{mol L}^{-1})^2$$

Simplify right-hand side:

$$k \times \text{mol}^3 \text{L}^{-3}$$

Step 3: Solve for units of k

$$k = \frac{\text{mol L}^{-1} \text{s}^{-1}}{\text{mol}^3 \text{L}^{-3}}$$

$$k = \text{mol}^{-2} \text{L}^2 \text{s}^{-1}$$



Final Answer:

$$k = \text{mol}^{-2} \text{L}^2 \text{s}^{-1}$$

Correct option: D

Question25

For an elementary reaction, $X(g) \longrightarrow Y(g) + Z(g)$, the $t_{1/2}$ is 10 min. In what period of time would the

concentration of X be reduced to 10% of its original concentration?

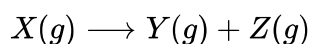
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Options:

- A. 20 min
- B. 33.2 min
- C. 15 min
- D. 25.2 min

Answer: B

Solution:



$$t_{1/2} = 10 \text{ min} \quad (\text{Given})$$

$$A = A_0 e^{-kt} \quad \dots\dots (i)$$

$$[A_0 = \text{Initial concentration}]$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{10} = 0.0693$$

From eq. (i),

$$0.1A_0 = A_0 e^{-0.0693 \times t}$$

$$\ln 0.1 = -0.0693 \times t$$

$$-2.303 = -0.0693t \Rightarrow t = 33.23 \text{ min.}$$

Question26

Which statement among the following is incorrect?

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Options:

- A. Unit of rate of reaction is ms^{-1} .

B. Unit of rate of disappearance is ms^{-1} .

C. Unit of rate constant k depends upon order of reaction.

D. Unit of rate constant k for a first order reaction is ms^{-1} .

Answer: D

Solution:

(b) Unit of rate of reaction $\frac{d[A]}{dt} = k[A]$

$$\text{Unit} = \text{ms}^{-1}$$

(a) Rate of disappearance

$$A \rightarrow B$$
$$-\frac{d[A]}{dt} \Rightarrow \text{ms}^{-1}$$

(d) For 1st order

$$\frac{-d[A]}{dt} = k[A]$$
$$\text{ms}^{-1} = k \text{ m}$$
$$k = \text{s}^{-1}$$

For 2nd order

$$\frac{-d[A]}{dt} = k[A]^2$$
$$\text{ms}^{-1} = k [\text{m}]^2$$
$$k = \text{m}^{-1} \text{s}^{-1}$$

So, k depends on order of reaction and unit of k for 1st order reaction is s^{-1} .

Question27

For zero order reaction, a plot of $t_{1/2}$ versus $[A]_0$ will be

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Options:

A. a straight line passing through the origin and slope = k

B. a horizontal line (parallel to x -axis)

C. a straight line with slope $-k$

D. a straight line passing through origin and slope $= \frac{1}{2k}$

Answer: D

Solution:

For zero order reaction,

$$[A] = [A_0] - kt \dots (i)$$

$$[A] = \frac{[A_0]}{2} \text{ (half-life) } \dots (ii)$$

From eqs. (i) and (ii)

$$\frac{[A_0]}{2} = [A_0] - kt_{1/2} \Rightarrow t_{1/2} = \frac{[A]_0}{2k}$$

Plot of $t_{1/2}$ versus $[A_0]$ will be straight line.

Question28

If the rate constant for a first order reaction is $2.303 \times 10^{-3} \text{ s}^{-1}$. Find the time required to reduce 4 g of the reactant to 0.2 g.

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Options:

A. 1.30 hours

B. 21.60 hours

C. 0.36 hours

D. 2.60 hours

Answer: C

Solution:

Initial weight (A_0) or a is 4 g

Final weight at time t or after reduce ($a - x$) or 0.2 g

First order reaction; $t = \frac{2.303}{k} \log \left(\frac{a}{a-x} \right)$

$$t = \frac{2.303}{2303 \times 10^{-3}} \log \left(\frac{49}{0.29} \right)$$

$$t = 10^3 \times \log 20$$

$$t = 1301 \text{ s or } t = 0.36 \text{ h}$$
